

Synthesis Paper

Predictive mapping of harmful algal blooms with Sentinel-2 imagery and random forest modelling

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ABSTRACT

Climate change and anthropogenic nutrient enrichment are driving the proliferation of algal blooms worldwide, posing a significant threat to aquatic ecosystems, including urban reservoirs. As water security in these systems is critical, inadequate monitoring continues to contribute to water quality deterioration and the loss of vital ecosystem services. This study aims to develop early detection tools for harmful algal blooms using Sentinel-2 multispectral imagery, the Random Forest algorithm, and in situ data. Model performance was evaluated for Trophic State Index (TSI), Chlorophyll-a (Chl-a), and surface water temperature in urban reservoirs, with initial calibration and validation on the A Baxe reservoir, followed by external validation in As Forcadas (northwest Spain). Spectral indices – such as the Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), Brightness Index (BI) – and specific bands were derived using Google Earth Engine (GEE). The developed models demonstrated high accuracy, with coefficients of determination (R^2) of 0.97 for Chl-a, > 0.80 for TSI, and > 0.70 for temperature. The TSI model's relevance is underscored by its integration of various physicochemical parameters (N_{total} , P_{total} , Secchi Disk, Chl-a), and the BI index showed strong performance in estimating temperature despite Sentinel-2's lack of a thermal band. This study proposes transferable methodologies for any reservoir, delivering an effective tool for the early detection of harmful algal blooms in stagnant waters, and contributing significantly to improved water resource management and the protection of aquatic ecosystems.

1. Introduction

Ensuring water quality in reservoirs is essential for promoting a healthy and sustainable society in harmony with the environment. Urban reservoirs are key for effective water management and urban sustainability [50]. In addition to supplying water, these reservoirs offer diverse ecosystem services, including hydroelectric power, recreation, fishing, and irrigation [19]. Furthermore, they can play a fundamental role as flood control systems during extreme weather events or as decentralized water supply systems in times of severe drought due to climate change [6]. However, pressure from agriculture, urban and industrial discharges, invasive species, land use changes, and climate-induced shifts in rainfall and temperature have negatively impacted water resources [47,59]. This has posed a significant challenge for local authorities, especially as regards providing quality water supply to urban populations [44].

Insufficient oversight of urban reservoirs has led to ecological degradation and water quality issues, including heavy metal contamination, biodiversity loss, fish kills, oxygen depletion, and eutrophication [53]. Specifically, eutrophication is a serious phenomenon caused by this excessive nutrient enrichment, leading to vigorous algae blooms and altering the structure of lakes and reservoirs [4]. One of the most problematic symptoms of excessive nutrient enrichment is the proliferation of harmful blooms of toxic cyanobacteria, known as CyanoHABs [67]. This phenomenon negatively impacts aquatic food production (fishing and aquaculture), recreational and tourism activities [105], and compromises drinking and irrigation water safety due to the release of cyanotoxins –compounds with unpleasant tastes and odours, surface foam formation, and oxygen depletion [54]. Consequently, CyanoHABs have caused disruptions to water supply and significant economic losses worldwide, in both developed and developing regions [20,52].

Therefore, the development of strategies and methodologies that

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provide managers with information on how to address, evaluate, and understand this phenomenon is essential to mitigate and reduce potential risks to the population. Accordingly, remote sensing emerges as a key tool for water security, allowing for the spatial mapping of surface blooms and assessment of their extent and severity [21]. The Sentinel-2A/B satellites (Copernicus program) offer high spatial resolution (10–20–60 m), ideal for monitoring inland waters and identifying small, heterogeneous algal patches [18,69]. Their 13 spectral bands, covering visible to shortwave infrared wavelengths and including the red-edge spectrum (B05–B07), make them particularly effective for assessing water quality in small reservoirs [2].

Many studies have used these satellite data with machine learning (ML) algorithms. However, most focus on a narrow set of variables, mainly chlorophyll-a (Chl-a), total suspended solids (TSS), Secchi Disk depth (SDD) and phycocyanin (PC). By contrast, we here sampled a broader set of in-situ variables – Chl-a, total phosphorus, total nitrogen, SDD, and surface water temperature (plus cyanobacteria counts) – and used them to derive the Trophic State Index (TSI) through machine learning modelling, a novel approach given that, to the best of our knowledge, Sentinel-2 data using predictive models have not been previously used to estimate TSI. This methodology not only enables the assessment of the trophic status of the reservoirs, but also provides spatially explicit maps to identify differences across areas within the water body. In addition, we modelled surface water temperature, a parameter that remains scarcely studied with Sentinel-2 imagery, which offers new opportunities for thermal monitoring in inland waters.

Several modelling approaches have been developed for the monitoring of CyanoHABs, ranging from data-driven techniques such as learning (DL) networks [98], to more traditional statistical models, including Multiple Linear Regression (MLR) [7,15], Multivariate Adaptive Regression Splines (MARS) [2,71] and Support Vector Regressions (SVR) [1,16]. While deep learning architectures generally require large datasets to avoid overfitting and ensure stable predictions, algorithms such as Random Forest are more robust with smaller sample sizes, making them well-suited for data-limited applications such as this study. Furthermore, other studies have investigated the relationship between individual spectral bands and remote sensing indices—such as the Normalized Difference Water Index (NDWI) [75,93], and the Normalized Difference Vegetation Index (NDVI) [95,98], — and key physicochemical parameters (e.g., Chl-a, TSS, Secchi depth) that determine water quality. These indices can provide rapid insights into bloom presence and extent; however, their performance is often limited when applied in optically complex waters or under varying environmental conditions, and they generally focus on one or two variables at a time.

Recent research has explored machine learning approaches capable of integrating multiple variables and spectral features simultaneously to overcome these limitations. Among these, Random Forest (RF) has emerged as a robust and interpretable model that performs well even with relatively small datasets [41], offering high predictive accuracy for multiple water quality indicators without the extensive data requirements typical of deep learning models [1,96].

Under this premise, the main objective of this study is to assess the performance of RF model when estimating the Trophic State Index (TSI), Chl-a, and surface temperature in urban reservoirs. These variables provide complementary insights into HAB dynamics: TSI integrates parameters such as Chl-a, phosphorus, nitrogen, and Secchi depth, offering a holistic view of nutrient enrichment and phytoplankton biomass, while surface water temperature – although scarcely explored with Sentinel-2 due to the absence of thermal bands – plays a key role in cyanobacterial growth rates, stratification, and seasonal bloom patterns. The integration of in-situ measurements with Sentinel-2 satellite data combines the spatially extensive and frequent coverage of remote sensing with the accuracy and specificity of field-based observations, improving model calibration, validation, and applicability to other water bodies. The models were calibrated and validated in the A Baxe reservoir (Galicia, NW Spain), a site that frequently experiences

CyanoHAB events. Other parameters measured in the field, such as total suspended solids, phycocyanin, or phycoerythrin, were not modelled here to maintain a scope focused on the selected key variables, but they remain relevant and will be considered in future applications of this methodology. Additionally, the developed models were validated in a second reservoir, As Forcadas (NW Spain), which shares similar issues but differs in specific pressures and characteristics. This cross-validation supports the generalizability of the proposed methodology. The tools and methods proposed here, relying on open-access platforms such as Sentinel-2, Google Earth Engine (GEE) and Python, provide a replicable and cost-effective framework. They contribute to integrated water resource management, aiding in the protection and sustainability of freshwater ecosystems.

2. Study area

The methodology was applied in two different scenarios: it was first calibrated and validated with data from the A Baxe reservoir, and then tested at the As Forcadas reservoir. A Baxe is located in the Autonomous Community of Galicia (NW Spain), within the Umia River Basin in the province of Pontevedra (Fig. 1). The Umia River rises in the Serra do Candán mountain range and flows through the Rías Baixas for a total of 70 km. In the year 2000, the Caldas de Reis dam created the A Baxe reservoir (6.15 km³) for water supply and hydroelectric purposes. The dam regulates river flows to supply the O Salnés region, maintain summer baseflows, and prevent flooding [93].

The region has a humid oceanic climate (Rías Baixas subtype), with high, regular rainfall and mild winters. In 2023, the mean annual temperature was ~ 15 °C, and precipitation averaged 5.8 L/m². Main land uses in the Umia Basin include hardwood forests (35.13 %), cultivated land (24.91 %), pastures and scrublands (15.48 %), and coniferous forests (10.07 %) [4,93].

On the other hand, the Forcadas reservoir, also in Galicia (Fig. 2), was built in 1966 on the Forcadas River (23 km), which drains into the Cedeira estuary. Located in Valdoviño (A Coruña), it has a capacity of 11 km³ and supplies water to the Ferrolterra region. Land cover includes hardwood forests (55.13 %), grasslands and heaths (17.63 %), cropland (11.21 %), and minimal conifers (0.01 %). In the same way as Caldas de Reis, the climate is mild and humid oceanic, with a 2023 average temperature of 14 °C and 5.9 L/m² precipitation. These factors are relevant, as cyanobacterial blooms intensify under warmer temperatures and reduced rainfall [62].

Both reservoirs, which are part of the Galicia-Coast Hydrographic Demarcation, have experienced recurrent cyanobacterial blooms, mainly of *Microcystis aeruginosa*, since 2006 [10].

3. Materials and methods

This methodology comprises two main stages (Fig. 3). First, satellite images are obtained via GEE, where imagery with the required specifications is downloaded, and multispectral values are extracted at the target coordinates for model training. Simultaneously, physicochemical data are processed to derive the TSI. These in-situ and satellite data are integrated to form the training dataset. A predictive model is then developed for each variable, evaluated using separate test data. Subsequently, the model is used to generate predictions – first over the training area, then applied to the second reservoir – resulting in raster maps with pixel-level variable estimates. These outputs are validated against in-situ measurements.

3.1. In-situ data

Field data were collected at two locations in A Baxe reservoir: upstream (UTM 531496E 4717366 N) and downstream (UTM 533539E 4716431 N), and one site in As Forcadas (UTM 573771E 4829616 N). Surface water samples, targeting microalgae concentrations, were

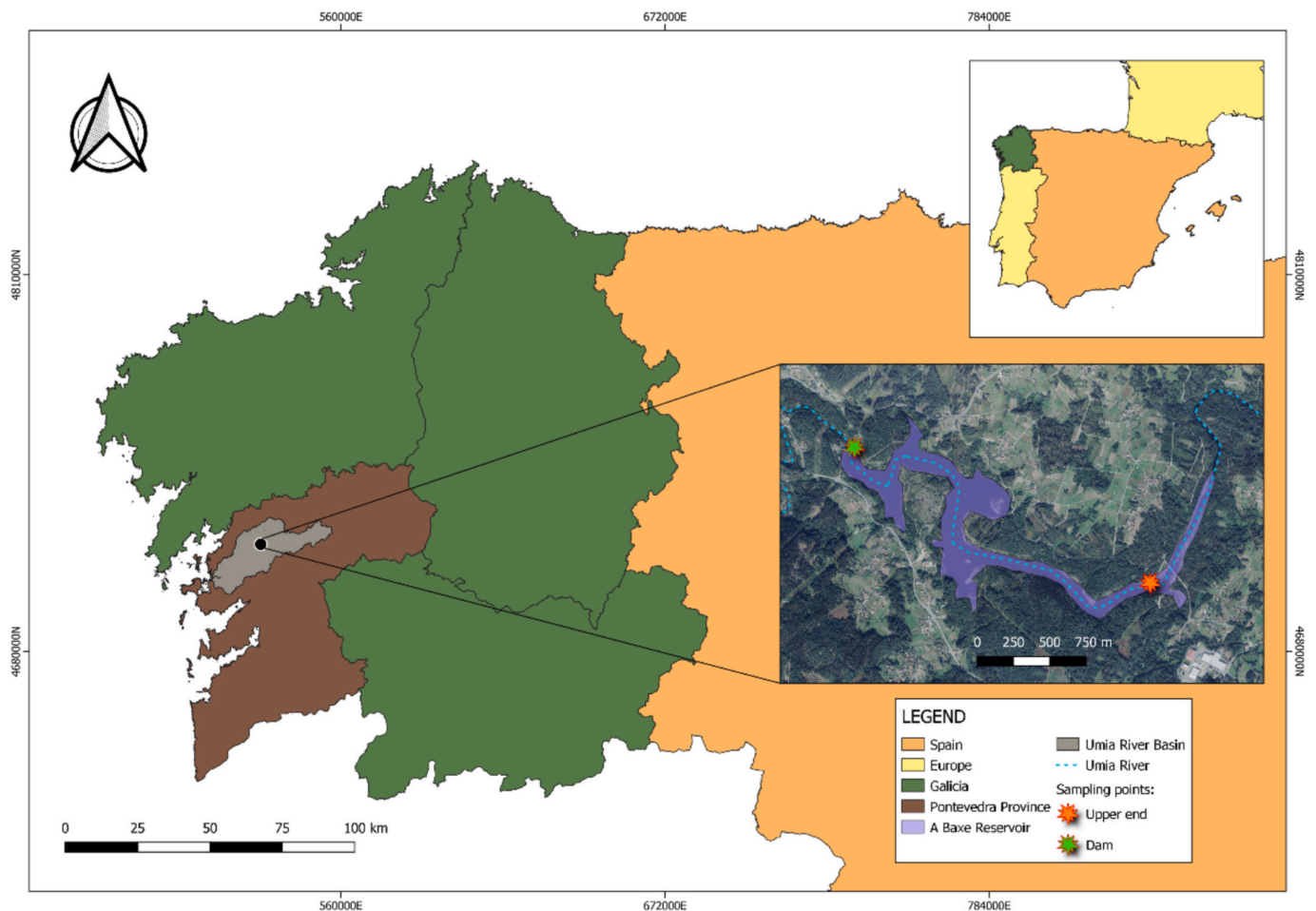


Fig. 1. Map of the study area showing the location of the A Baxe reservoir (EPSG: 25829).

collected as per UNE-EN 15,972 standards. Parameters measured included Chl-a (mg/L), total nitrogen and phosphorus, temperature, Secchi depth, and cyanobacteria abundance (cells/mL). Analyses were conducted using a desktop multiparameter spectrophotometer (Hanna Instruments, HI 33224).

3.2. Sentinel-2 data preprocessing

Sentinel-2 imagery is provided at two processing levels: Level 1C (Top of Atmosphere, TOA) and Level 2A (Bottom of Atmosphere, BOA) reflectance [27]. In this study, Level 2A images were used, accessed via the Google Earth Engine (GEE) API, specifically through the harmonized S2_SR_HARMONIZED dataset. GEE is a JavaScript-based open platform that enables cloud-based processing of global-scale remote sensing data, improving efficiency and accessibility [12].

Imagery was selected to match in-situ data dates or up to three days apart when cloud-free images (<2% cloud cover) were unavailable. A secondary filter using the QA60 band excluded images with clouds over sampling points, ensuring reliable multispectral data and minimizing atmospheric interference. Table 1 lists the in-situ sampling dates and corresponding imagery used for model training. The full dataset, including discarded observations due to outliers or cloud cover, is available on Zenodo[3].

The use of GEE streamlined image acquisition, enabling a raster combining all bands of interest for the study area to be directly downloaded. Sentinel-2 L2A bands at 10 m (B02, B03, B04) and 20 m (B05) were used. In GEE, the 20-m band was automatically resampled to 10 m using bilinear interpolation to allow pixel-wise combination with the 10-m bands. All band values were subsequently normalized to surface

reflectance (0–1) by applying the quantification factor (10,000) (Table 1A). This involves defining the polygon geometry to spatially filter the image collection, and applying additional filters by date, spectral bands, and cloud cover.

Multispectral values were normalized using the Quantification Value (10,000) via the GEE API and used to generate a.csv dataset for model training. This dataset includes one column per band and one per in-situ variable, with rows representing each sampling date and location. As for the division of the data, 80 % was used for training and the remaining 20 % for testing.

3.3. Indices computed from Sentinel-2 bands

The Normalized Difference Water Index (NDWI distinguishes surface water using near-infrared (NIR) and green bands, exploiting water's low reflectance due to strong absorption of electromagnetic radiation[34]. NDWI values range from −1 (dry) to 1 (water), with surface water typically between 0.2 and 1.0 [14]. It was calculated using Eq. (1) within the same environment used to obtain the satellite imagery. In this study, NDWI was applied during the image processing stage to segment the water body before applying the predictive models, ensuring that estimations were restricted to water pixels and excluding land or other non-water surfaces. A threshold value > 0.2 was used to classify pixels as water. This segmentation was consistently applied in both the calibration/validation phase for the A Baxe reservoir and in the transferability analysis for the As Forcadas reservoir.

$$NDWI = \frac{Green - NIR}{Green + NIR} \quad (1)$$

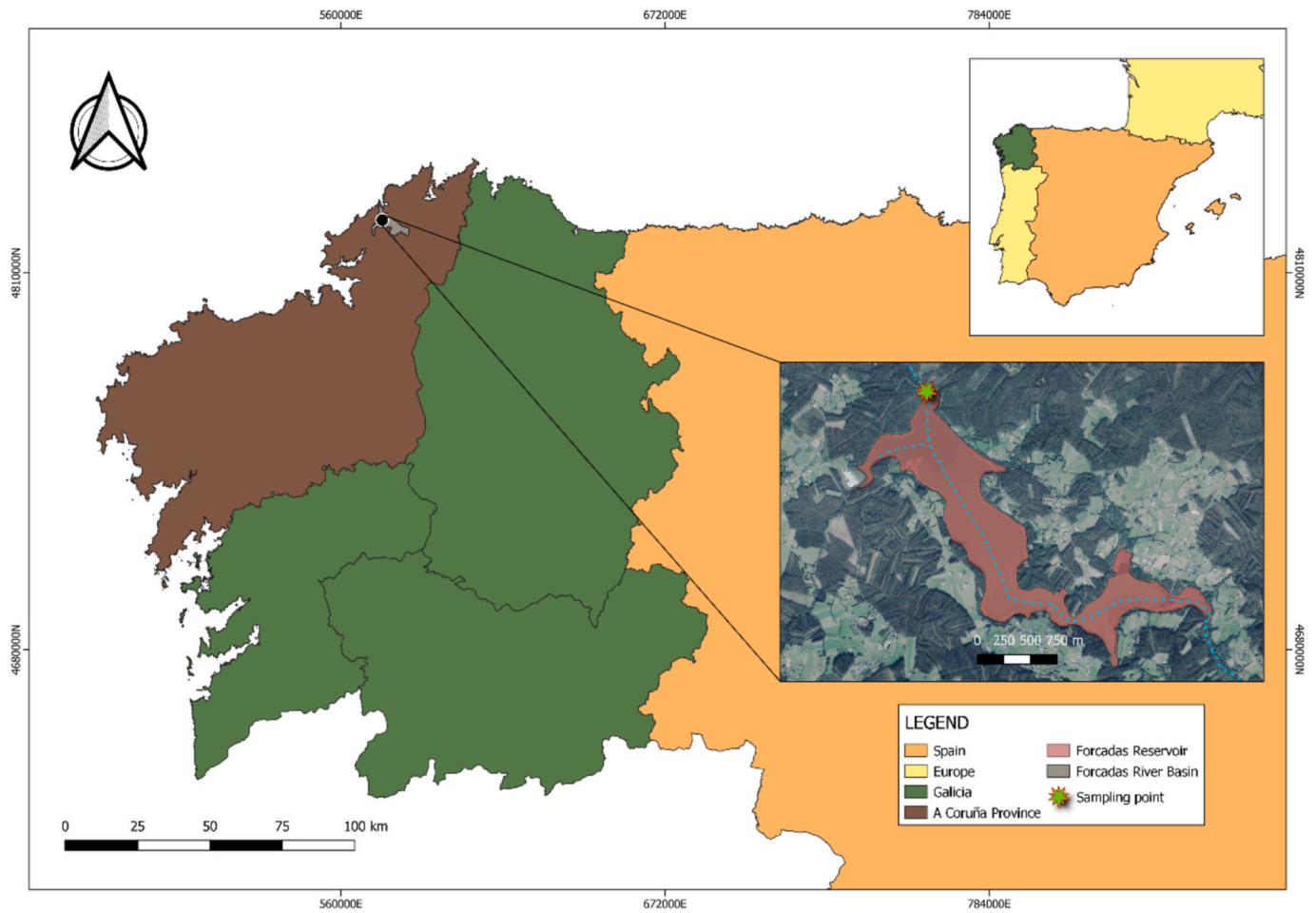


Fig. 2. Map of the study area showing the location of the Forcadas reservoir (EPSG: 25829).

The Normalized Difference Chlorophyll Index (NDCI) estimates chlorophyll in turbid waters using red and red-edge bands [55]. Values close to 1 indicate high chlorophyll, while negative values indicate absence. This index is also computed automatically during image processing (Eq. (2)).

$$NDCI = \frac{Red\ Edge - Red}{Red\ Edge + Red} \quad (2)$$

El Brightness Index (BI) represents the average brightness of a satellite image. This index is sensitive to the brightness of soils which is highly correlated to the humidity and presence of salts in surface [30,57] (Eq.3).

$$BI = \sqrt{((Red * Red) + (Green * Green)) / 2} \quad (3)$$

3.4. Determining of trophic State: Trophic state Index (TSI)

The study area's trophic state was estimated using the TSI developed by Carlson [23] and modified by Zhang et al. [101]. TSI was calculated via logarithmic transformations of in-situ measurements of Chl-a, total phosphorus (P_{total}), total nitrogen (N_{total}), and Secchi Disk depth (DS) (Eqs. 4–8):

$$TSI(Chl - a) = 10 \times (2.5 + 1.086 \times \ln(Chl - a)) \quad (4)$$

$$TSI(P_{total}) = 10 \times (9.436 + 1.624 \times \ln(P_{total})) \quad (5)$$

$$TSI(N_{total}) = 10 \times (5.453 + 1.694 \times \ln(N_{total})) \quad (6)$$

$$TSI(SD) = 10 \times (5.118 - 1.94 \times \ln(DS)) \quad (7)$$

$$TSI_{index} = (TSI(Chl - a) + TSI(P_{total}) + TSI(N_{total}) + TSI(SD)) / 4 \quad (8)$$

This index categorizes trophic status as oligotrophic, mesotrophic, eutrophic, or hypertrophic (Table 2).

3.5. Machine learning for PC modelling CyanoHABs

The ML algorithm selected was RF, an ensemble method effective for classification and regression [17]. It has been widely applied to monitor water quality using satellite data [42]. RF constructs multiple regression trees by bootstrapping n samples from the original dataset [28,74]. Additionally, it has been demonstrated that RF performs optimally when working with limited data [37,70].

A Random Forest Regressor was trained using Python (version 3.10.13) and the scikit-learn library, with the pre-processed dataset. RF is suitable for regression tasks and allows hyperparameter tuning to enhance model performance. Given the limited number of sampling points (two in A Baxe and one in As Forcadas), a conventional 80/20 split was applied to ensure that both training and validation sets contained representative data while preserving as much information as possible for model training. More advanced validation schemes, such as leave-one-date-out cross-validation, were considered; however, the temporal distribution of observations, the small number of coincident in situ-satellite acquisitions meeting quality requirements (minimal cloud cover and suitable atmospheric conditions), and the limited temporal overlap between sites were found to compromise the statistical robustness of the resulting performance metrics.

A random search algorithm was applied to optimize the model, testing combinations of hyperparameters and selecting the one with highest accuracy. The following hyperparameters were evaluated [17,35,84]:

- n estimators: The number of trees
- max depth: The maximum depth of the tree

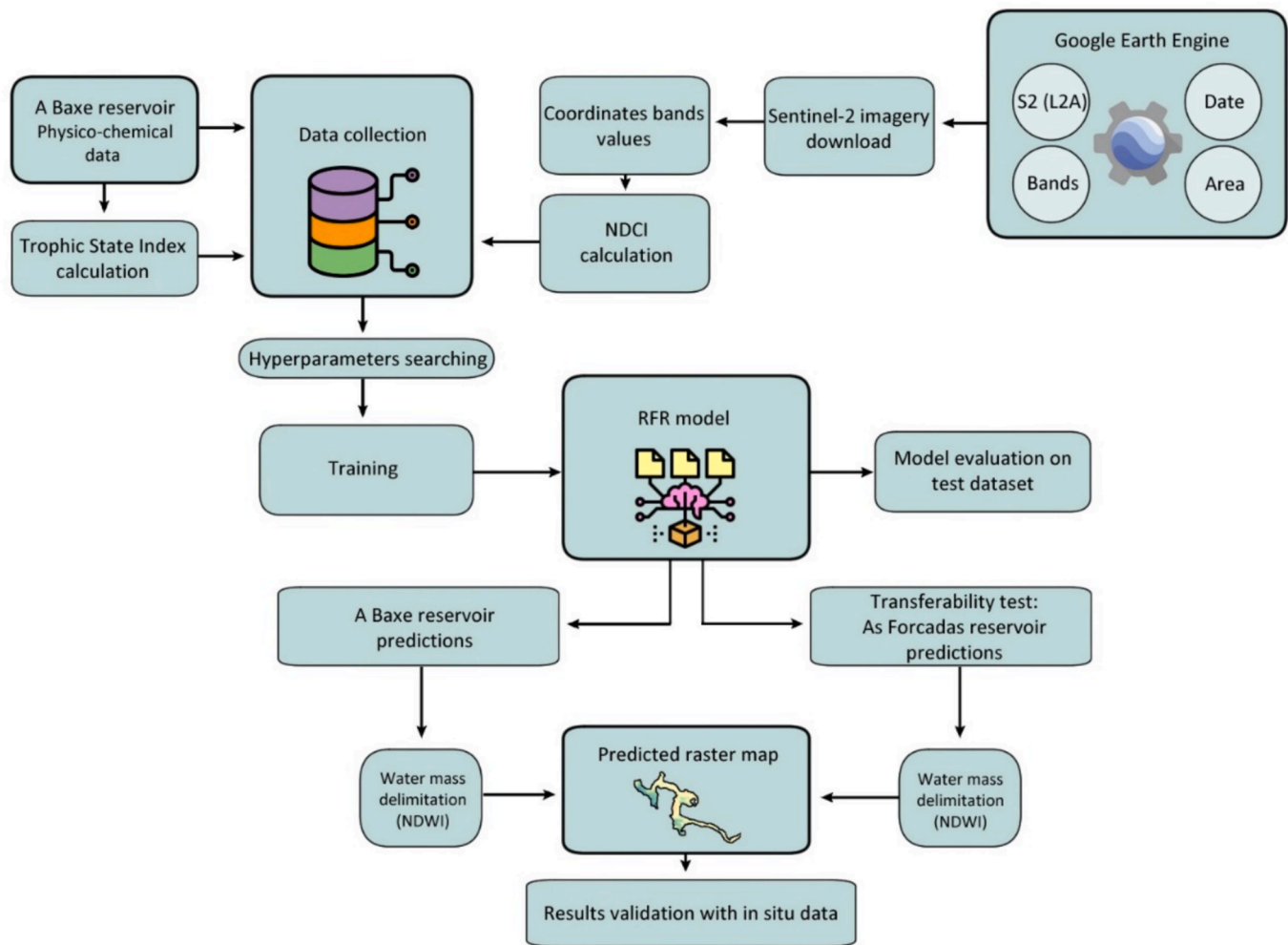


Fig. 3. Workflow for the developed technique, which employs Random Forest as a machine learning method and Sentinel-2 for assessing various variables related to eutrophication.

Table 1A
Characteristics of the Sentinel-2 bands used in the methodology developed.

Band	Central Wavelength (nm)	Resolution (m)
B01 (aerosol)	443	60
B02 (blue)	490	10
B03 (green)	560	10
B04 (red)	665	10
B05 (red edge)	705	20
B06	740	20
B07	783	20
B08 (NIR)	842	10
B08A	865	20
B09	940	60
B10	1375	60
B11 (SWIR 1)	1610	20
B12 (SWIR 2)	2190	20

- min samples split: The minimum number of samples required to split an internal node
- min samples leaf: The minimum number of samples required to be at a leaf node

The hyperparameter search space included 81 possible combinations, as detailed in Table 3.

3.5.1. Optimized models for water quality parameters

The following models were optimized to predict different water quality parameters: TSI, Chl-a, and surface temperature. Input features for each model were selected based on their correlation to in-situ data, and the RF hyperparameters were optimized as described in the previous section.

Table 4 shows the hyperparameters for each model along with the input data used. These models were implemented and trained using the scikit-learn library.

These optimized models were used to generate raster layers where each pixel represents the predicted value for the respective water quality parameters.

3.6. Validation metrics

Models were evaluated using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Coefficient of Determination (R^2) metrics. The R^2 measures the proportion of variance explained by the model, ranging from 0 to 1, where 1 indicates a perfect fit and 0 indicates a poor fit. R^2 values above 0.7 are considered good, between 0.5 and 0.7 are moderate, and below 0.5 are poor.

The MAE quantifies the average absolute deviation between predicted and observed values, as expressed in Eq. 9:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9)$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value and n is the number of data samples.

Table 1
Dates of in situ samples and Sentinel-2 images.

TSI		Chl-a		Temperature	
In situ	Image	In situ	Image	In situ	Image
20/06/2017	17/06/2017	09/10/2017	07/10/2017	14/06/2017	14/06/2017
14/08/2017	13/08/2017	27/08/2018	26/08/2018	04/10/2017	02/10/2017
09/10/2017	12/10/2017	22/10/2018	22/10/2018	08/08/2018	08/08/2018
26/02/2018	24/02/2018	13/05/2019	13/05/2019	03/10/2018	02/10/2018
27/08/2018	26/08/2018	19/08/2019	21/08/2019	07/08/2019	08/08/2019
22/10/2018	22/10/2018	16/09/2019	15/09/2019	01/07/2020	03/07/2020
07/01/2019	05/01/2019	02/12/2019	04/12/2019	28/09/2020	29/09/2020
13/05/2019	13/05/2019	15/03/2021	15/03/2021	01/09/2021	30/08/2021
19/08/2019	21/08/2019	07/06/2021	06/06/2021	09/06/2022	11/06/2022
16/09/2019	15/09/2019	01/09/2021	30/08/2021	03/08/2022	31/08/2022
02/12/2019	04/12/2019	15/03/2022	13/03/2022	06/09/2022	06/09/2022
03/08/2020	05/08/2020	09/06/2022	11/06/2022	15/03/2023	15/03/2023
13/10/2020	16/10/2020	03/08/2022	31/07/2022	11/05/2023	09/05/2023
07/06/2021	08/06/2021	06/09/2022	06/09/2022		

Table 2
TSI ranges definition.

Trophic State	TSI RANGE
Hypertrophic	>70
Eutrophic	51–70
Mesotrophic	40–50
Oligotrophic	<40

Table 3
Hyperparameters values for model optimization.

Hyperparameter	Value 1	Value 2	Value 3
n_estimators	50	100	200
max_depth	None	10	20
min_samples_split	2	5	10
min_samples_leaf	1	2	4

Table 4
Optimized hyperparameters for each model.

Parameter	Trophic State Index (TSI)	Chlorophyll-a (Chl-a)	Temperature
Input Features	B03, B04, B05, NDCI	B02, B03, B04, B05	Brightness Index (BI)
n_estimators	100	100	200
max_depth	20	10	10
min_samples_split	2	2	5
min_samples_leaf	1	1	1
Output	Raster of predicted TSI values	Raster of predicted Chl-a values	Raster of predicted temperature values

The RMSE calculates the square root of the average squared differences between predicted and actual values, highlighting significant deviations, as defined in Eq. 10:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

where $\hat{y}_i$ is the predicted value, y_i is the actual value, and n is the number of data samples.

3.7. Model development and transferability

After evaluating the model on the test dataset, its transferability was assessed by processing images from the As Forcadas reservoir. The model automatically segments the water mass using NDWI to focus predictions on the area of interest, selects the necessary bands for the prediction, and processes a new raster with the optimal features for estimating the variable. The final output is a segmented raster where each pixel represents the estimated variable value. Transferability was evaluated here by applying the Random Forest models trained in the A Baxe reservoir directly to the As Forcadas reservoir without re-training. This approach tests whether the models can generalize across two sites with similar climate and typology but different specific pressures, demonstrating the potential for applying the methodology in other locations where in-situ data are scarce or unavailable, relying solely on remotely sensed predictors.

4. Results and Discussion

4.1. Calibration and validation of the TSI model in the A Baxe reservoir

As a result of the application of the developed TSI-model, a complete and detailed representation of the trophic state of the reservoir during the periods analysed was obtained (Fig. 4). From a statistical perspective, an R^2 of 0.71 was achieved, indicating a high degree of fit for the analysed data (G. [77]). Furthermore, the model developed for the TSI obtained an RMSE of 2.61 and an MAE of 2.12, indicating a low difference between the observed values and those predicted by the model. This set of statistical measures demonstrates that the results conform to the observed reality, proving the effectiveness of the developed model. To date, few studies have mapped TSI using satellite imagery. Studies evaluating this trophic index typically consider only the chlorophyll variable for its calculation [29,49,103,104], which partially limits the model's accuracy in reflecting the true ground conditions of the study area. In this case, by taking a broader set of variables such as Chl-a, P_{total} , N_{total} , and Secchi depth into account, a more comprehensive insight into the trophic state of the reservoir is expected to be obtained. On the other hand, Shang et al., [85] found statistical differences when mapping parameters such as Chromophoric Dissolved Organic Matter (CDOM), with better results in oligotrophic conditions ($R^2 = 0.86$, RMSE = 0.07) compared to eutrophic ones ($R^2 = 0.82$, RMSE = 3.64). Due to the limited sample size, statistical analyses were conducted for combined trophic states.

On the one hand, the Green (B03), Red (B04), and Red Edge 1 (B05) bands, as well as the NDCI index, were used in this model. Shang et al., [85] also found the B05/B03 ratio optimal for trophic index evaluation. In our case, the NDCI index, which is part of the TSI calculation (Eq. 4), has already demonstrated a high predictive capacity for Chl-a estimation in our study area ($R^2 = 0.97$, RMSE = 7.83 $\mu\text{g/L}$, MAE = 6.54 $\mu\text{g/L}$; see Section 4.2), which supports its inclusion as a key variable for TSI modelling. Red, green, blue, and NIR bands are commonly used for nitrogen and phosphorus analysis [22,39]. Kislik et al. [43] found a strong correlation to Band 02, though it was not detected in this study.

In October 2017 (Fig. 4A), the reservoir's trophic state remained mesotrophic, with upper ends and margins nearing eutrophic values (Fig. 4B). This difference is due to decreased flow and water movement along the banks, increasing nutrient accumulation and sunlight availability, which worsens the trophic state [62]. Shallower water allows greater sunlight penetration and raises surface water temperature, promoting microalgae growth [24,88], and nutrient absorption [66].

In contrast, the August 2020 image (Fig. 4C) shows higher TSI values (Fig. 4D), with the entire reservoir close to eutrophic status. This trend, especially in summer, is due to the favourable conditions for

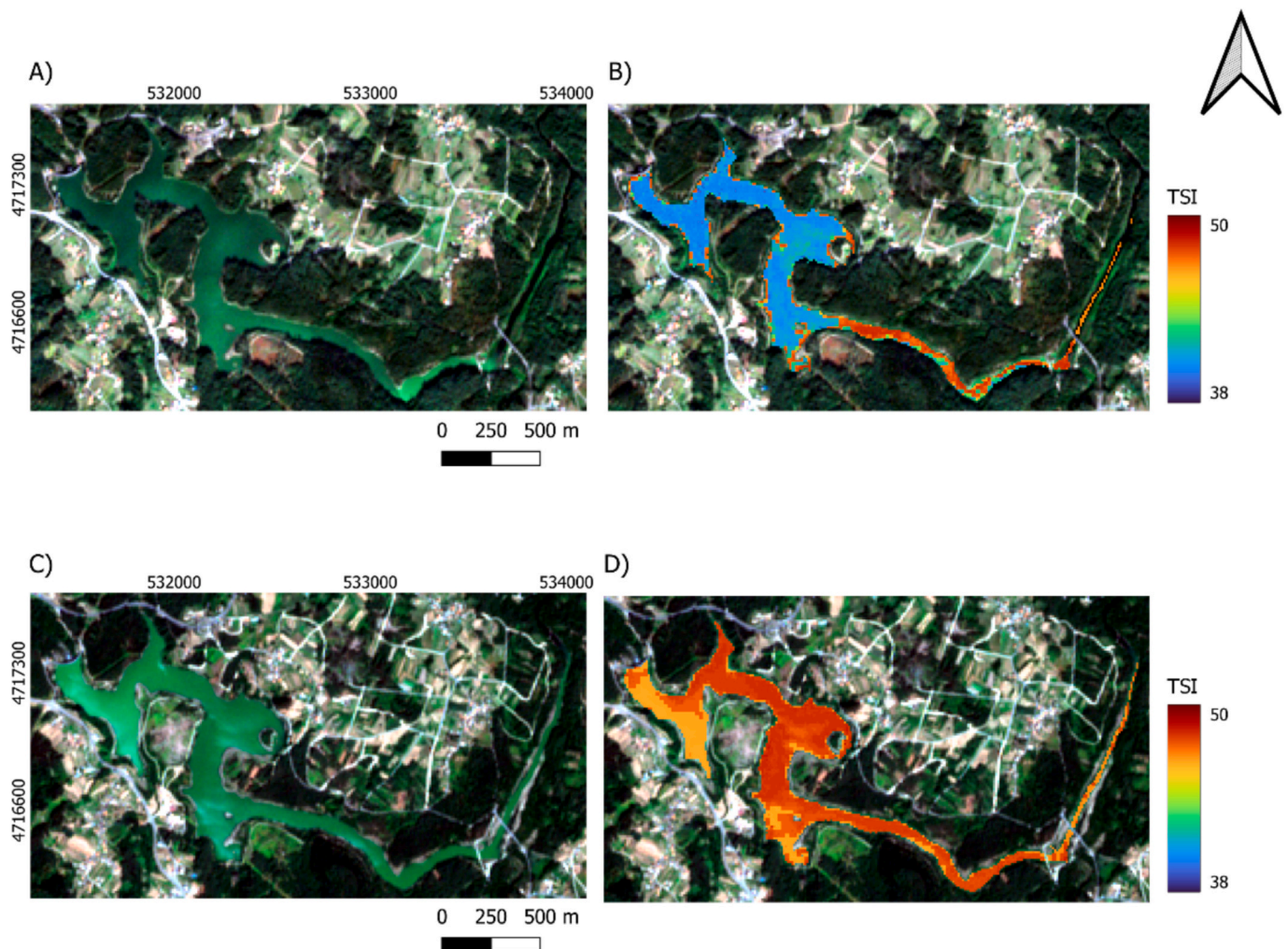


Fig. 4. Model developed for TSI in the A Baxe reservoir area. (A) August 2020 S2 image (low TSI value), (B) TSI prediction (August 2020), (C) October 2017 S2 image (high TSI value), (D) TSI prediction (October 2017).

cyanobacteria growth. Increased water temperature strengthens thermal stratification, limiting dissolved oxygen and biogeochemical processes [31,87], while high temperatures are a key factor in cyanobacterial blooms [40] and reduced water transparency [65], both factors included in the TSI calculation.

4.2. Calibration and validation of the Chl-a model in the A Baxe reservoir

An R^2 of 0.97 was achieved for the Chl-a model (Fig. 5), indicating a very high fit. However, the RMSE and MAE values were 7.83 and 6.54, respectively, reflecting some discrepancy between observed and predicted data. This discrepancy is mainly due to the inherent variability in Chl-a data, with significant peaks and high median values [48]. The model's performance is slightly superior to other studies. For example, Wang et al., [94] reported an R^2 of 0.68 and RMSE of 10.16 for a Chl-a model based on RF, and Yajima & Derot, [97] achieved an R^2 of 0.52 for lakes in Japan. Than et al., [90] found an R^2 of 0.57 using RF. Other machine learning methods, such as CatBoost [100], reported an R^2 of 0.84 and RMSE of 5.12 $\mu\text{g/L}$, and Chen et al. [25], achieved an R^2 of 0.83 with SVM. The RF method delivers favourable results with limited data and shorter processing times.

The model's successful operation required comprehensive data pre-processing, including the removal of outliers and data affected by cloud cover. Careful selection of Sentinel-2 bands was also crucial. Viso-Vázquez et al., [93] found a high correlation to B03 for Chl-a ($R^2 =$

0.80). Salgado et al [82] highlighted the spectral range of 530 to 600 nm as indicative of suspended solids, a key factor in microalgae blooms. Saberioon et al., [80] found good correlation to B05 ($R^2 = 0.59$) but a negative correlation to B03 and B04. However, Ha et al., [36] reported better correlation to B03 than B04, emphasizing the influence of biogeochemical factors such as algae, dissolved organic matter, and suspended solids. Kutser et al., [45] found a strong correlation to B05 ($R^2 = 0.83$). Kislik et al., [43] studied NDCI and found correlations ranging from $R^2 = 0.43$ to 0.84 across different reservoirs. Various factors influence Chl-a correlations, such as microalgae composition, reservoir biogeochemistry, and spectral band characteristics [93]. The pixel size is also significant, especially for smaller reservoirs [13].

In October 2017 (Fig. 5A), Chl-a concentrations were around 15 $\mu\text{g/L}$ (Fig. 5B), with higher levels at the reservoir's upper end, reaching 90 $\mu\text{g/L}$. This decrease is linked to higher precipitation, which enhances scouring and destratification [73], leading to a reduction in cyanobacteria. This results in increased transport of nutrients and sediments towards the downstream end, promoting accumulation.

The model also performs well under high Chl-a conditions, as seen in August 2020 (Fig. 5C), with concentrations of 50–60 $\mu\text{g/L}$ (Fig. 5D). In areas near the dam, levels exceeded 100 $\mu\text{g/L}$, supported by elevated nutrient concentrations, temperature, and solar radiation. Stagnation during dry periods [38] further increases cyanobacteria and microalgae concentrations, stimulating chlorophyll production [8]. Higher chlorophyll levels directly impact dissolved oxygen, with increases linked to

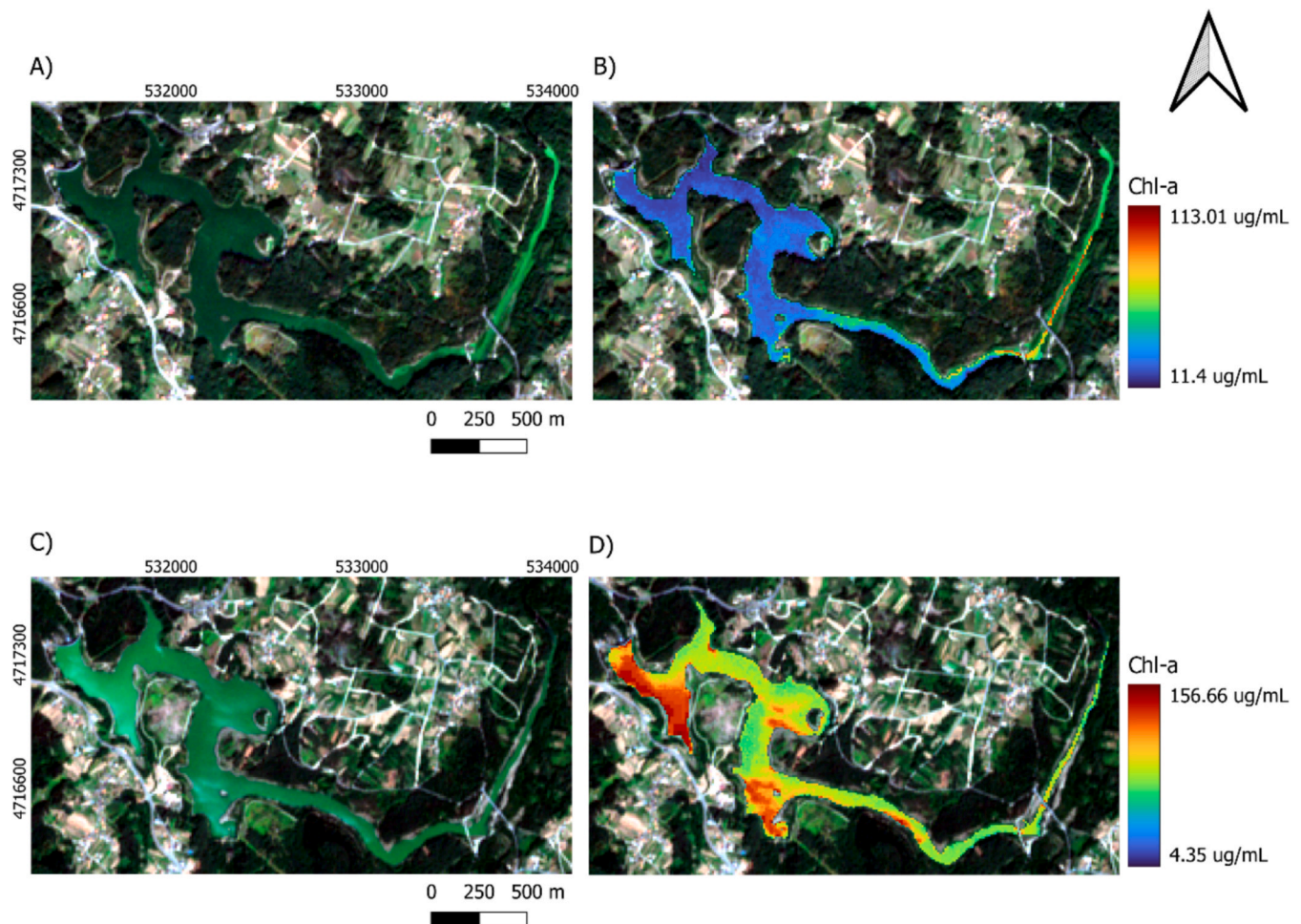


Fig. 5. Model developed for Chl-a in the A Baxe reservoir area. (A) October 2017 S2 image (low Chl-a concentration), (B) Chl-a prediction (October 2017), (C) August 2020 S2 image (high Chl-a concentration), (D) Chl-a prediction (August 2020).

lower oxygen levels, thus degrading water quality [58].

4.3. Calibration and validation of the temperature model in the A Baxe reservoir

A final model was developed where water temperature was studied (Fig. 6). The model achieved an R^2 of 0.80, indicating a high degree of correlation. The RMSE and MAE values were 1.91 and 1.51, respectively, indicating a low discrepancy between measured and predicted values. Few studies have used Sentinel-2 for surface water temperature in reservoirs due to the lack of a thermal band [33], unlike Sentinel-3, which has a thermal band but a larger pixel size, unsuitable for this study area [76]. Some studies, such as Nasser et al. [60], used Sentinel-2 for land temperature with an R^2 of 0.90, though the methodology overestimated the water surface temperature in this study.

The use of the BI index (B03 and B04) proved valuable for assessing surface water temperature due to its sensitivity to surface brightness variations, which capture changes in water temperature. Higher water temperatures often decrease surface brightness due to changes in water reflectance and absorption [86]. Additionally, BI is influenced by humidity and salts, which are also important variables [61]. The observed correlation between BI and water temperature suggests this index is a reliable indicator for estimating water temperature from satellite imagery.

In March 2023 (Fig. 6A), the reservoir's temperature was around 12 °C (Fig. 6B), contrasting with in situ data. Lower temperatures reduce cyanobacteria concentrations and increase dissolved oxygen. When

temperatures are low in winter and spring, dissolved oxygen levels are high; higher temperatures in summer and autumn mean lower oxygen levels. In August 2018 (Fig. 6C), temperatures ranged from 20–25 °C (Fig. 6D). Higher temperatures in summer favour cyanobacterial blooms, promoting photosynthesis and reproduction. Temperatures above 20 °C accelerate cyanobacterial development [79], while increased temperatures also enhance water column stratification, facilitating cyanobacterial growth [83]. Several species reach their maximum growth rate at temperatures above 25 °C [65,83]. Additionally, some areas experience lower temperatures due to shading from riparian vegetation, which reduces solar radiation and softens the temperature [11].

4.4. Model testing at the As Forcadas reservoir

After testing in the study area, the model was applied to predict the same variables in the As Forcadas Reservoir (Fig. 7). Satellite images and in situ data were used to validate the predictions.

An image from May 2021 (Fig. 7A) was processed to predict TSI. The predicted value was 49.26 (Fig. 7B), compared to the in situ value of 55.38. Although there is a slight difference, the prediction provides a fairly accurate estimate, especially considering the model was not trained with data from this reservoir or similar systems. The same image was processed to estimate Chl-a concentration (Fig. 7C), yielding a predicted value of 34.27, compared to the in situ value of 23.89. This margin is acceptable given the variability in Chl-a concentrations. As other studies suggest, several factors influence the correlation between

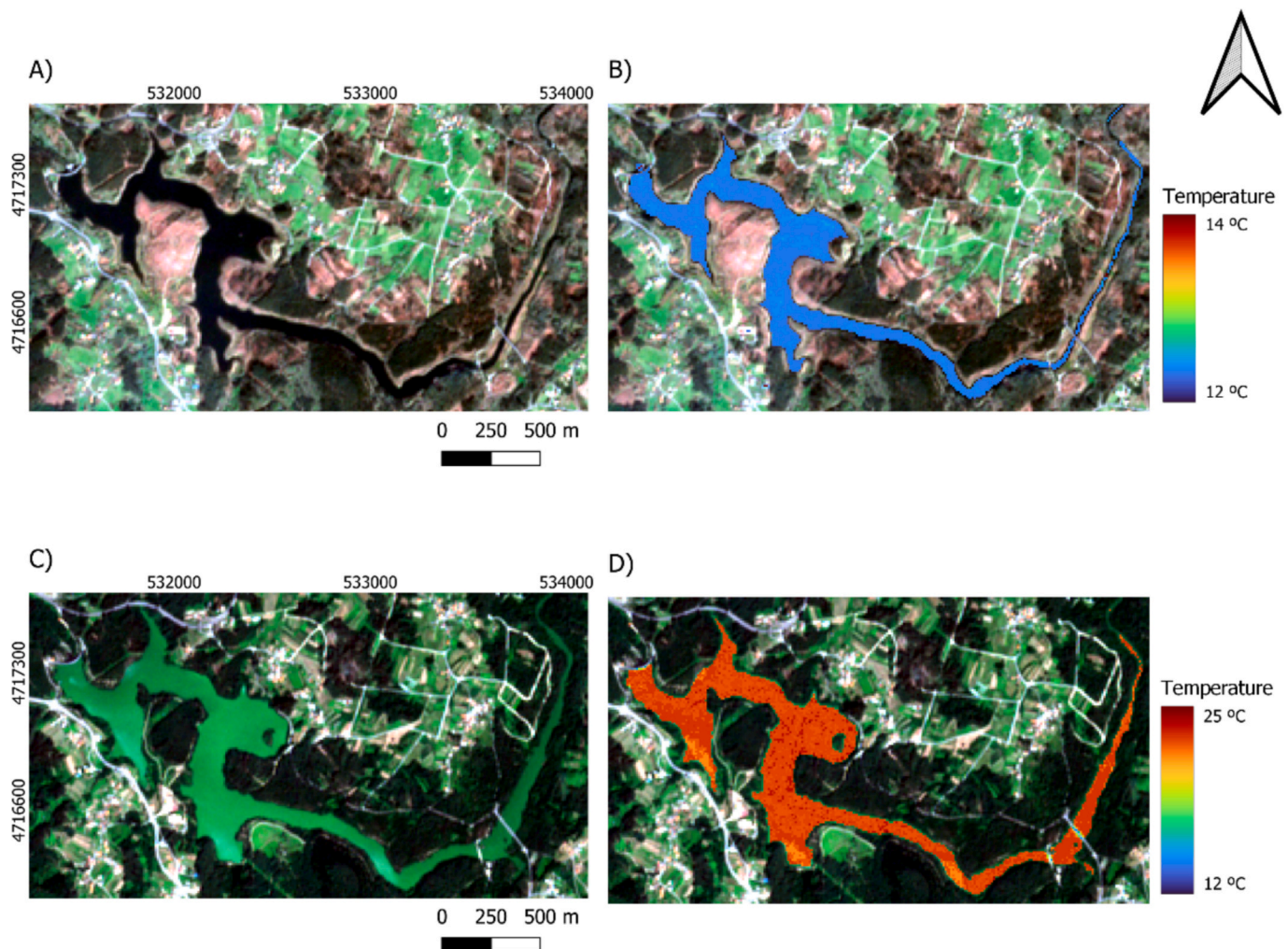


Fig. 6. Model developed for temperature in the A Baxe reservoir area. (A) March 2023 S2 image (low temperature), (B) Temperature prediction (March 2023), (C) August 2018 S2 image (high temperature), (D) Temperature prediction (August 2018).

Chl-a and spectral bands, such as microalgal species composition, water column height, and sediment presence[93].

As regards temperature (Fig. 7D), the predicted value was 18.6 °C, compared to the in situ value of 16.5 °C. This small difference is minimal, considering the satellite pass timing and the absence of the Sentinel-2 thermal band. It is worth noting that the lower part of the reservoir, where no eutrophication issues were observed, exhibited lower temperatures, TSI values, and Chl-a concentrations.

4.5. General Discussion

The developed models showed that the chlorophyll model achieved the highest R^2 (0.97). The models for TSI (R^2 0.71) and temperature (R^2 0.80) balanced both metrics, demonstrating strong predictive capacity. These results outperform deep learning-based algorithms. For instance, Lee et al. [46] developed an MLP model for Chl-a prediction in a Korean river with an R^2 of 0.86, and Zhao et al. [102] used a CNN to predict Chl-a with an R^2 of 0.79. D. Zhang et al. [99] achieved an R^2 of 0.66 for TSI prediction using a DNN. These findings underscore the effectiveness of machine learning models in limited sample scenarios, as opposed to deep learning, which requires large datasets for reliable results [89]. Although these studies were conducted in different regions, with varying datasets and modelling approaches, the reported metrics provide a useful contextual reference to assess the relative magnitude of our results.

However, some limitations must be addressed, such as the limited input data, Sentinel-2 pixel size[21] and few sampling points (two in A Baxe, one in As Forcadas)[93], which may restrict spatial variability capture. In total, 25 surface water samples were collected between 2017 and 2023 in the A Baxe reservoir. In addition, one sampling point was included in the As Forcadas reservoir to provide external validation of the model. While this sample size is limited, it is sufficient for the Random Forest methodology, which performs robustly with small datasets and allows reliable modelling even when data availability is constrained. A formal evaluation of data-augmentation effects (e.g., training models with and without augmentation) was not conducted due to the limited sample size, which could lead to unstable results; this will be explored in future work with larger datasets.

The evaluation in a single external location (As Forcadas) provides an initial indication of the method's potential for transferability, but does not allow for a conclusive validation of its generalizability. Further testing in additional reservoirs, with different optical properties and trophic conditions, will be required to confirm the method's applicability across diverse settings. In addition, feature ablation studies could provide valuable insights into the contribution of each spectral band and index to model performance. Although not included in this study due to the small dataset size and potential instability in performance metrics when features are removed, such analyses will be considered in future research with expanded datasets.

Additionally, the availability of autumn–winter images is limited due

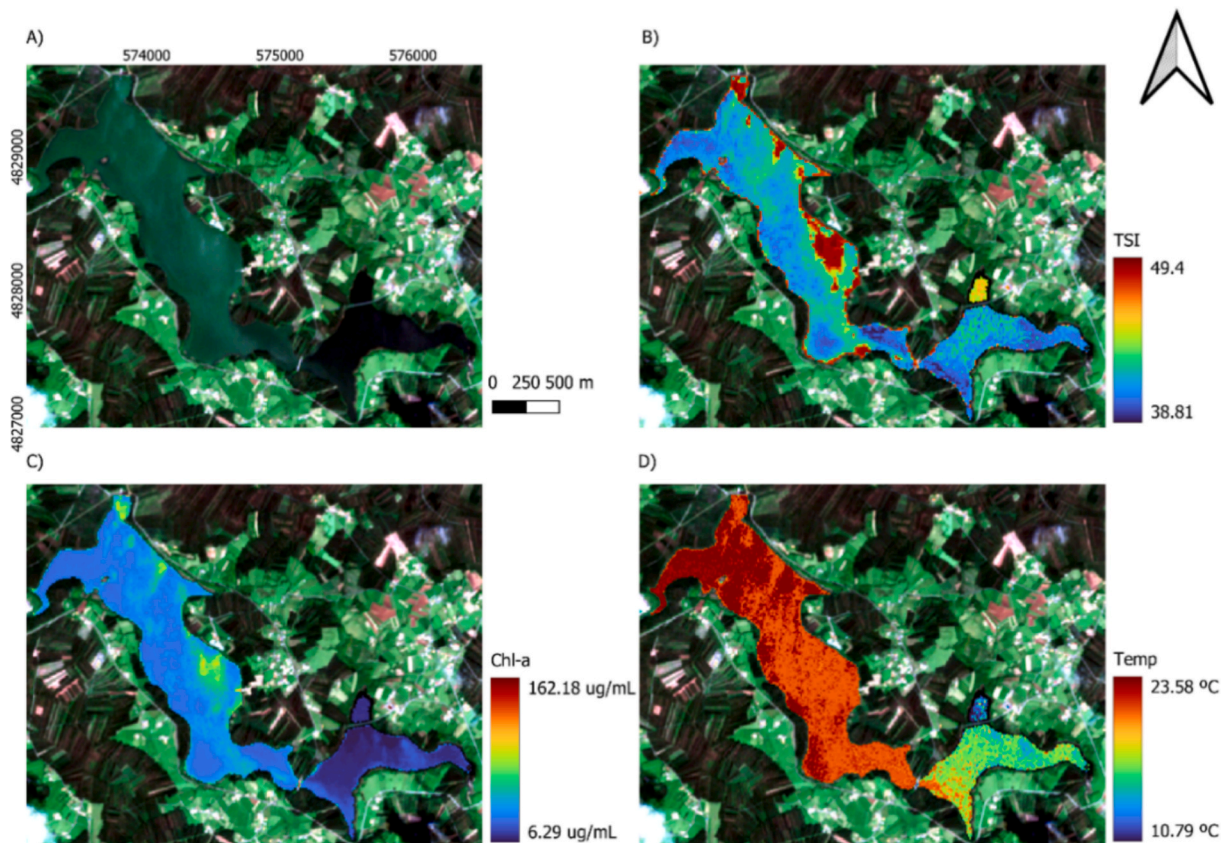


Fig. 7. Models developed in the As Forcadas area. (A) May 2021 S2 image, (B) TSI prediction (May 2021), (C) Chl-a prediction (May 2021), (D) Temperature prediction (May 2021).

to cloud cover, affecting model training during low cyanobacterial activity periods.

The RF model, evaluated in limited-sample studies[63,78], performed well in this study, especially for TSI, which can be challenging due to its trophic status variability[85]. The surface temperature analysis, complicated by the absence of a thermal band in Sentinel-2, yielded good results using the BI index. This index, previously used for soil moisture analysis[92], shows promise for water temperature estimation. Other satellites, such as Landsat-8 or Sentinel-3, cannot be effectively applied due to low spatial resolution.

These methodologies for mapping trophic state, Chl-a, and water temperature are crucial for developing early warning systems. Accurate knowledge of lentic system status, using satellite imagery and machine learning, can enhance decision-making and contribute to sustainable water management[5]. Climate change is expected to increase these risks[68], making such tools vital for managing water supply and strengthening hydraulic system resilience[56]. The models tested at As Forcadas could be adapted to other regions, taking interfering variables into consideration[72]. Although this study focused on the Random Forest algorithm due to its robustness with limited datasets and interpretability, future research could compare its performance with other machine learning techniques such as Decision Trees, Artificial Neural Networks, and Support Vector Machines to further assess predictive accuracy for harmful algal bloom mapping.

On the other hand, the need for improvement in watershed management and planning should be emphasized. This environmental governance of the watershed must be based on these powerful tools, improving multi-criteria decision-making and allowing multi-objective optimization of the solutions. In this way, the different risks faced by lentic systems will be addressed simultaneously to products and drivers of change within the global socio-ecological system, thus enhancing the

understanding of freshwater in complex, multidisciplinary, and multi-objective decision-making[9]. To achieve this, it will be essential to address the various issues impacting these ecosystems, such as changes in land cover[91] or degradation of riparian vegetation[51]. These impacts should be addressed through reforestation of abandoned areas [64], improvement of riparian vegetation[81], or better application of agricultural treatments [26]. Additionally, the use of these tools will enable anticipation of adverse effects of climate change and action to avoid or minimize expected damages[32]. In conclusion, the set of models studied here will be key to ensure the water security of lentic systems, in particular to assess the increasingly frequent issue of eutrophication.

5. Conclusions

The evaluation of the Random Forest machine learning algorithm, in combination with Sentinel-2 satellite imagery, was evaluated for the estimation and spatial mapping of the Trophic State Index (TSI), chlorophyll-a (Chl-a), and surface water temperature in two urban reservoirs. The models were initially calibrated and validated for the A Baxe reservoir (Galicia, northwestern Spain), and subsequently assessed for their transferability to the As Forcadas reservoir, located in the same region. This study offers a novel contribution by conducting a comprehensive analysis of the TSI, considering the full set of physicochemical parameters that comprise it –namely, total nitrogen (N_{total}), total phosphorus (P_{total}), Secchi disk depth (SD), and Chl-a. Additionally, the potential of the Sentinel-2 sensor for evaluating the surface water temperature was explored, despite its lack of a thermal band. Notably, the Brightness Index (BI) demonstrated strong performance in temperature estimation, suggesting a promising and previously unexplored approach for monitoring this variable. The results obtained support the

effectiveness of these machine learning tools in evaluating lentic systems. The TSI model showed a value of R^2 of 0.71 (RMSE 2.61, MAE 2.12), while the chlorophyll model achieved an R^2 of 0.97 (RMSE 7.83, MAE 6.54), and the surface water temperature model obtained an R^2 of 0.80 (RMSE 1.91, MAE 1.51). The methodology used in this study provides an effective tool for mapping various variables associated with the eutrophication of lakes and reservoirs, thereby improving multicriteria decision-making and enabling multi-objective optimization of the solutions to be adopted.

CRedit authorship contribution statement

Carolina Acuña-Alonso: Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Formal analysis, Conceptualization. **Diego Barba-Barragans:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joaquín Martínez-Sánchez:** Supervision, Project administration, Funding acquisition. **Xana Álvarez:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.wasec.2025.100194>.

Data availability

We have shared data on the models in the following link: <https://doi.org/10.5281/zenodo.15045568>

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